

Q9 Ans: C

For an object of mass m orbiting around Earth,
Gravitational force provides for centripetal force

$$G \frac{M_{\text{Earth}} m}{r^2} = \frac{mv^2}{r}$$

$$GM_{\text{Earth}} \left(\frac{1}{r} \right) = v^2$$

Since they are both orbiting around same Earth,

$$\left(\frac{1}{r} \right) \propto v^2$$

$$\left(\frac{v_{\text{ISS}}}{v_{\text{satellite}}} \right)^2 = \frac{r_{\text{satellite}}}{r_{\text{ISS}}}$$

$$v_{\text{ISS}} = v_{\text{satellite}} \sqrt{\left(\frac{r_{\text{satellite}}}{r_{\text{ISS}}} \right)} = 3.1 \sqrt{\left(\frac{4.2 \times 10^4}{6.8 \times 10^3} \right)} = 7.7 \text{ km s}^{-1}$$



